**SUMMER TERM 2023**
**CENTRALLY-MANAGED ONLINE EXAMINATION**
**ECON0039: ADVANCED MACROECONOMICS**

**Time allowance**
You have 2 hours to complete this examination, plus additional collation time of 20 minutes and an Upload Window of 20 minutes. The additional collation time has been provided to cover any additional tasks that may be required when collating documents for upload, and the Upload Window is for uploading, completing the Cover Sheet and correcting any minor mistakes. The additional collation time and Upload Window time allowance should not be used for additional writing time.

If you have been granted SoRA extra time and/or rest breaks, your individual examination duration and additional collation time will be extended pro-rata and you will also have the 20-minute Upload Window added to your individual duration.

All work must be submitted anonymously in a PDF file and you should follow the instructions for submitting an online examination in the AssessmentUCL Guidance for Students.

If you miss the submission deadline by up to 40 minutes late in the Late Submission Period, a late submission penalty will be applied unless you submit a valid claim for Technical Failures. At the end of the 40-minute Late Submission Period, you will not be able to submit your work via AssessmentUCL and you will not be permitted to submit the work via email or any other channel. If you are unable to submit your work or have submitted your work late during the Late Submission Period due to technical difficulties which are substantial and beyond your control, you should submit a claim for Technical Failures via the AssessmentUCL Query Form.

**Page Limit:** No page limit

**Query on the Examination Paper:** : If you have a query about the examination paper, instructions or rubric, you should complete an AssessmentUCL Query Form. Please note that you will not receive a response during your examination.

**Academic Misconduct:** By submitting this assessment, you are confirming that you have not violated UCL's Assessment Regulations relating to Academic Misconduct contained in Section 9 of Chapter 6 of the Academic Manual.

**All questions should be answered.**

**Part A carries 40% of the total mark, Part B 20% and Part C 40%.**

## PART A : The Liquidity Trap

THIS problem considers a general equilibrium model of the liquidity trap. We assume an endowment economy with two periods (plus a continuation value). The two periods are denoted 1 and 2. The representative household enters period one with an amount of money $M_0$ and an amount of bonds $B_0 = 0$. They receive endowment $y_1$ in period one and $y_2$ in period two. Money is the numéraire. Preferences are given by

$$u(c_1) + \beta u(c_2) + \beta^2 V(M_2, B_2)$$

where $V$ is the continuation value of holding $M_2$ units of money and $B_2$ units of bonds at the end of period 2. It is assumed that $u' > 0$, $v'_M > 0$ and $v'_B > 0$. We also make the assumption that $0 < \beta < 1$.

In each period $t$, the household must first decide on the quantity of bonds $B_t$ to issue (and repay next period) and on the quantity of money $M_t$ to hold. The nominal interest rate on bonds issued in period $t$ is $i_t$. After this decision has been made, the household decides on its consumption (price $P_t$), that has to be bought with money $M_t$ (consuming one's own endowment is forbidden) according to the cash-in-advance constraint:

$$P_t c_t \leq M_t$$

It also sells its endowment at price $p_t$. The set of constraints that the representative household faces therefore contains two budget constraints and two cash-in-advance constraints. The budget constraints are

$$\begin{array}{rcll} P_1 c_1 + M_1 + B_1 & \leq & P_1 y_1 + M_0 + X_1, & (\lambda_1) \\ P_2 c_2 + M_2 + B_2 & \leq & P_2 y_2 + M_1 + X_2 + (1 + i_1) B_1, & (\lambda_2) \end{array}$$

where $X_t$ is new money injection that is distributed to the household at the beginning of the period. $\lambda_t$ is the Lagrangian multiplier associated to the budget constraint of period $t$. The cash-in-advance constraints are

$$\begin{array}{rcll} P_1 c_1 & \leq & M_1, & (\mu_1) \\ P_2 c_2 & \leq & M_2, & (\mu_2) \end{array}$$

where $\mu_t$ is the Lagrangian multiplier associated to the cash-in-advance constraint of period $t$.

**A.1** Write the Lagrangian of the household problem and first order conditions of the maximisation, including complementary slackness ones.

**A.2** Explain what is the purpose of assuming a continuation value $V(M_2, B_2)$ for $M_2$ and $B_2$.

**A.3** Prove $\lambda_1 > 0$, $\lambda_2 > 0$. Also prove that $\mu_1 > 0$ when $i_1 > 0$. What is the economic interpretation of this last result?

**A.4** Define a general equilibrium of this economy. Explain why the bond market equilibrium condition is $B_t = 0$.

Let's make the assumption that $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ with $\sigma \geq 0$, that $V(M_2, B_2) = \log(M_2) + \log(1+B_2)$ and that $M_2 > 1$.

**A.5** Solve for period 2 equilibrium and find the equilibrium values of $(\lambda_2, \mu_2, c_2, P_2)$. What is the equilibrium effect in period 2 of monetary expansion $X_2 > 0$? Discuss.

**A.6** Solve for period 1 equilibrium $(\lambda_1, \mu_1, c_1, P_1, i_1)$. What is the condition on $M_1 = M_0 + X_1$ under which the economy is in the liquidity trap? Explain.

Assume now that $\sigma = 1$, $y_1 = 1$, $M_0 = \frac{1}{\beta}$ and $X_1 = 0$. Prices are flexible in period 2, but fixed in period 1, at the level $P_1 = \frac{1}{\beta^3}$.

**A.7** Show that $c_1 < y_1$ and explain why. Collapse the period 1 equilibrium condition into two equations in $c_1$ and $P_1$, given the equilibrium in period 2. Make a graph and show that expansionary monetary policy (to $M_1 = \frac{1}{\beta^2}$) can increase $c_1$ to the level $\beta$. Show that increasing $M_1$ further cannot lead to $c_1 = y_1 = 1$. Explain why.

## PART B : Question

WHY ARE the positive co-movements between output, investment and consumption a challenge for business cycle modelling? Write a structured short essay of no more than 500 words (less is fine).

## PART C : Discussion

BELOW IS an article published by Cardiff Garcia on the Financial Times Alphaville blog in July 2017 and entitled "When Keynes pondered Malthus". Read carefully this article (you can skip the endnotes) before answering the questions below. For each question, you should propose a structured answer.

C.1 Write a 10-line summary.

C.2 Explain the functioning of the Malthusian regime. Why has the economy exited that regime? (No more than 500 words, less is fine)

C.3 What was the dynamics of income inequalities according to Ricardo in the Malthusian regime? How have inequalities evolved since? (No more than 500 words, less is fine)

"When Keynes pondered Malthus "
by Cardiff Garcia
*Financial Times Alphaville blog*, Jul 26th 2017

In a conversation with Clifford Geertz in 1976, the economist Albert O Hirschman playfully expressed the first law of the social sciences: "Whenever a phenomenon in the social world is fully explained, it ceases to operate."

Perhaps the best historical example of this idea, or at least the most famous in the history of economic thought, was the publication of An Essay on the Principle of Population by Thomas Malthus in 1798.

Paul Krugman blogs that "over the roughly 60 centuries that have passed since civilization emerged in Mesopotamia, the Malthusian proposition – population pressure swallows up any gains in productivity, so that most people live on the edge of subsistence – was true for 58. It just so happens that the two centuries for which the proposition didn't hold were the two centuries after Malthus wrote."[1]

The nineteenth century would provide more such examples of economic thinkers having been right in their time, but wrong for all the time since.

By the estimates of economic historian Robert Allen, average output per capita did start rising around the year 1800. The technologies of the industrial age were coming on stream. But real wage growth stayed flat until roughly 1840, just before Engels published The Condition of the Working Class in England in 1844. Based on his firsthand observations in Manchester, Engels wrote that the English middle classes "prefer to ignore the distress of the workers and this is particularly true of the industrialists, who grow rich on the misery of the mass of wage earners".

Until that point, Engels was right about the nineteenth century – and he wasn't alone. According to Allen:

> Ricardo, Malthus, and Marx all believed that real wages would remain constant during capitalist development. They differed, however, in their explanations: Ricardo and Malthus believed that population growth would accelerate in response to any rise in income and ultimately force wages back to subsistence; Marx, on the other hand, believed that technological progress had a labour saving bias that would eliminate any upward demand pressure on wages even as output per worker surged.

They would all be proved wrong about the events that came after them. Marx and Engels were immediately wrong following their publication of The Communist Manifesto in 1848, just as real wages were starting to catch up to productivity growth.

Allen writes, my emphasis:

> The acceleration of productivity growth did, indeed, shift income from workers to capitalists, as [Marx] expected. The result, however, was not continually increasing immiseration, for the capitalists invested a portion of their extra income and the increase in the capital stock eventually allowed rising productivity to be manifest as rising real wages. History did, indeed, exhibit a stage pattern of evolution, but *the stage of flat real wages was followed by the most sustained rise in real wages ever seen* - not by socialist revolution.

Real wage growth caught up to productivity growth by 1860, exceeding it thereafter. “The ‘modern’ pattern was established”, Allen writes.

Krugman cites examples even earlier than Malthus, asking:

> Is there a moral to this story? Maybe it is that things economic do change. By and large they change more slowly than many people think; understanding the 1930s was still immensely valuable to understanding the world after 2008. But economic ‘laws’ – I generally hate that term – aren’t immutable, and good economists can be both right about the past and wrong for today.

True – though of course, paradigmatic shifts as dramatic as the end of the Malthusian reality are exceedingly rare. Does the ongoing transition to a world in which automation replaces even heavily cognitive tasks represent another?[2]

We should not be surprised that the question remains hotly debated and unanswerable. Historically it has often taken a very long time to confirm that the world had indeed changed. The example of Malthus is again instructive.

To use Hirschman’s formulation, the social phenomenon explained by Malthus in his essay – that a rising population would always overwhelm increased output – did cease to operate at very nearly the same time that the essay was published. But the cessation wasn’t clear to economic observers at the time. It would be contested throughout the nineteenth century and in fact was not definitively accepted even twelve decades later.

John Maynard Keynes published The Economic Consequences of the Peace, his brutal excoriation of the Paris Peace Conference, in 1919. The first and second chapters of Consequences explain the socioeconomic trends that defined Europe in the half-century before the start of the Great War. In the essay Keynes is brilliantly alive to the extraordinary, revolutionary nature of economic growth in the last third of the nineteenth century.

His very first lines warned against taking for granted its permanence:

> The power to become habituated to his surroundings is a marked characteristic of mankind. ... We assume some of the most peculiar and temporary of our late advantages as natural, permanent, and to be depended on, and we lay our plans accordingly. On this sandy and false foundation we scheme for social improvement and dress our political platforms, pursue our animosities and particular ambitions, and feel ourselves with enough margin in hand to foster, not assuage, civil conflict in the European family.

Consequences was not primarily about the continued relevance of Malthusian economics. Its focus was the punitive economic demands imposed by the Allied victors on the defeated Germans and, according to Keynes, the existential threat to European stability posed by these demands.[3]

But the "sandy and false foundation" – the assumption of rising living standards on which Europeans had come to rely before the War – also included the possibility that the economic forces observed since 1870 represented merely an anomalous and temporary deviation.

Keynes even suspected that the world had already started returning to its flatter Malthusian trajectory by the turn of the century:

> The pressure of population on food, which had already been balanced by the accessibility of supplies from America, became for the first time in recorded history definitely reversed. As numbers increased, food was actually easier to secure. ... Up to about 1900 a unit of labor applied to industry yielded year by a year a purchasing power over an increasing quantity of food. It is possible that about the year 1900 this process began to be reversed, and a diminishing yield of Nature to man's effort was beginning to reassert itself.

Keynes understood that the world into which he was born was an "economic Eldorado". His contemporaries risked forgetting the prior, gloomier eras that came before theirs:

> Before the eighteenth century mankind entertained no false hopes. To lay the illusions which grew popular at that age's latter end, Malthus disclosed a Devil. For half a century all serious economical writings held that Devil in clear prospect. For the next half century he was chained up and out of sight. Now perhaps we have loosed him again.

Now perhaps we have loosed him again. By then, approximately 120 years had passed since the "devil" had first been chained – and the finest economic mind of the time (and perhaps of any time) still did not know if the chains would be strong enough to hold him.

Modern economic observers should recognise the next big economic mutation more quickly than was ever possible for their predecessors. The social sciences are much further developed as disciplines than in the time of Keynes, much less the time of Malthus and Ricardo, much less that of Hume and Smith. The development of useful national accounts would not start for another decade after the publication of Consequences in 1919.

Keynes wasn't completely in the dark when he wrote Consequences. He had access to rudimentary but useful data about foreign trade and investment, coal production, crop yields, and population numbers for the relevant countries. He wrote about the number and kind of Germany's mercantile marine vessels ceded to the Allies. In the final chapter he encouraged the Allied countries to fully cancel the debts they owed each other, presenting a simple chart.

But there was nothing like the virtual armies of social scientists now devoting their careers to understanding trends that would be considered impossibly granular in the 1920s. Not only in the academy, but embedded within the statistical agencies of every reputable government and multilateral institution.

Plenty of confusion still exists about current topics, even – or especially – about the defiantly enigmatic issue of productivity growth. Its fluctuations remain hard to measure, hard to understand, impossible to predict.

But if a new, unexpected, and previously unimagined economic dynamic does assert itself, it seems unlikely that another twelve decades will pass before economic thinkers realise that the world has been knocked off its post-Malthusian trajectory and onto a new path.

## Notes

[1]Democratic Athens and a few other, earlier societies might well have breached the Malthusian cap, but just temporarily. Krugman's point holds.

[2]I first made similar points in the different context of the looming robot takeover, and my attention was directed to the example of Keynes on Malthus in Economic Consequences by Brad DeLong. Brad writes:

> When we come to the 1950s of Simon Kuznets, Robert Solow, and Walt Whitman Rostow, the Malthusian Devil is ignored as a result of the confluence of the demographic transition, the technological creativity of the industrial research lab, and the political victory of social democracy: output per worker is seen as high and growing indefinitely; moreover, income inequality is under control-a Dickensian (or Marxist) society of great wealth but also great mass poverty is off the table as well.

[3]Keynes worried about more than just a newly Malthusian existence. He also warned against the immediate return to the freewheeling laissez faire liberal-capitalism of the late nineteenth century, preferring lenient terms for Germany and debt forgiveness for the net-debtor Allies. For a critical assessment of Economic Consequences and a helpful treatment of the power dynamics contributing to the terms of peace, try chapter 15 of Adam Tooze's The Deluge, from which:

> Generations of economists have picked apart the flaws in Keynes's argument. But his critique stands both as a reflection and a contributor to the mood of disillusionment that followed Versailles. Combining the authority of an economic expert, the colour of an insider and magnificent rhetorical force,

> Keynes's book sold in the hundreds of thousands. It was quoted verbatim in the Republican assault on Wilson's treaty in the US Senate. Both Lenin and Trotsky recommended Keynes as essential reading to the Comintern. He was welcomed with open arms in Germany and helped to poison yet further the atmosphere between London and Paris. Superficially, of course, Keynes stood on Germany's side.
>
> But it was far from obvious, even from a German standpoint, whether he did not do more harm than good. His influence encouraged those in Germany who insisted that any payment was impossible, when a good faith effort to honour the treaty, even if it had fallen short, might well have steered the Weimar Republic away from the ruinous crisis of 1923. Of course, the point is not that Keynes was personally responsible for the ensuing disaster. The point is that rather than reading Keynes as a guide to the reparations problem, we should see The Economic Consequences as symptomatic of the crisis that is the central preoccupation of this book [The Deluge]. ...

But also:

> ... Keynes's aim was not to to create an elaborate long-term structure of government control to replace free trade or private lending. But he was scathing about those who believed that 'with the early removal of obstacles in the form of the blockage and similar measures to free international intercourse, private enterprise may be safely entrusted with the task of finding the solution'. ... The governments of Europe and America must act to restore basic lines of credit so that private initiative might then take over, otherwise those countries most in need of credit would be caught in a vicious circle of economic crisis, political uncertainty and diminished creditworthiness.